

# CARSON SLATER

+1(480) 320-8898 ♦ Waco, TX

[carsonslater7@gmail.com](mailto:carsonslater7@gmail.com) ♦ [in/carsonslateratbaylor](https://in/carsonslateratbaylor) ♦ <https://carsonslater.github.io>

## EDUCATION

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**PhD Statistical Science**, Baylor University 2024 - 2027 [Expected]  
Research involves developing statistical methods for time series clustering with applications in urban water distribution processes.

**M.S. Statistical Science**, Baylor University 2023 - 2024  
(GPA: 3.86/4.0)

**B.S. Applied Mathematics with Statistics**, Wheaton College 2019 - 2023

**B.A. Economics**  
(GPA: 3.46/4.0), *Edman Presidential Scholarship*

## EXPERIENCE

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**Research Assistant** January 2024 - Present  
Baylor University Waco, TX

- Developing statistical methods for urban water distribution processes, leveraging unsupervised learning on time series for customer-level water utility data.
- Collaborated with domain experts in urban water demand planning to comprehend data and processes, perform robust analyses, and effectively communicate results.
- Developed R Shiny dashboard to enable collaborators at Denver Water to visualize the outdoor irrigation trends in homes with advanced metering infrastructure.
- Edited and reviewed material for an undergraduate intro to data science course. ([DOI: 10.7910/DVN/BZJW2E](https://doi.org/10.7910/DVN/BZJW2E))

**Graduate Assistant** August 2023 - December 2025  
Baylor University Waco, TX

- Consulted for ongoing interdisciplinary research in the computer engineering and environmental science departments, and for the American Holistic Nursing Association. Published academic projects in peer-reviewed journals.
- Delivered quality reports for each client and had regular meetings to discuss the project, the methodology and results.
- Teaching assistant for Computational Statistics (R).

**Academic Coach** January - May 2023  
Wheaton Academy Center for Lifelong Learning West Chicago, IL

- Mentored students 1-on-1 to sharpen math skills from Algebra I to AP Statistics.

**Mathematics Teaching Assistant** August 2022 - May 2023  
Wheaton College Wheaton, IL

- Facilitated help sessions for undergraduate students in the topics of Machine Learning and Probability Theory.

## SKILLS

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| <b>Technology</b>     | Docker, Positron ( $\approx$ VS Code), Git/Github, reveal.js, R/RStudio, SQL, duckDB, $\LaTeX$                                                                                                                                                                                                                                                                                                                            |
| <b>Python</b>         | pandas, numpy, PyTorch (Lightning), plotnine, seaborn, plotly, scikit-learn, Jupyter                                                                                                                                                                                                                                                                                                                                      |
| <b>R</b>              | tidyverse (dplyr, ggplot2, etc.), TensorFlow (Keras), torch, shiny, RMarkdown/Quarto, rstan, rjags, caret, tidymodels, targets (gittargets, proffer, etc.)                                                                                                                                                                                                                                                                |
| <b>Stats &amp; ML</b> | statistical modeling, hypothesis testing, mixture models, time series, regression, linear mixed models, clustering, classification, deep learning, PCA, random forests, ensemble models (e.g., XGBoost), SVM, Bayesian methods, spatiotemporal modeling                                                                                                                                                                   |
| <b>Coursework</b>     | Statistical Bioinformatics, Deep Learning, Bayesian Methods, Bayesian Theory, Survival Analysis, Time Series, Theory of Statistics, Statistical Methods, Computational Statistics, Numerical Analysis, Linear Algebra for Data Analysis, Mathematical Statistics, Machine Learning, Probability Theory, Real Analysis, Intensive Introduction to Computer Science (Harvard), Econometrics, Economics of Labor and Poverty |